Centralized Audit Trail Compliance Engine

NAME: SAGAR K **SRN**: PES2UG24CS424

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall parse structured JSON audit logs and identify and mask sensitive PII fields such as Social Security Numbers and credit card numbers before persisting records to storage.	High	Stored audit logs contain masked PII and a SHA-256 integrity hash.	Protects sensitive personal information and supports GDPR compliance.	Given requirement
FR-002	Functional	The system shall receive and validate audit logs submitted by enterprise applications.	High	A valid JSON audit log is accepted and processed.	Ensures that only valid audit events enter the compliance system.	Validation should reject malformed JSON.
FR-003	Functional	The system shall generate and store a SHA-256 integrity hash for every processed audit log.	High	Every persisted audit record contains a SHA-256 hash and recomputing the hash produces the same value.	Provides cryptographic evidence that audit records have not been modified.	Hash must be generated after PII masking.
FR-004	Functional	The system shall allow a Compliance Officer to search and filter audit logs using fields such as date, user, event type, and severity.	High	A valid filter returns all matching audit records without exposing unmasked PII.	Enables compliance officers to investigate and review audit activity efficiently.	Filters should support multiple conditions.
FR-005	Functional	The system shall allow a Security Auditor to export selected audit records as a cryptographically verifiable immutable audit report.	High	The exported report contains the selected masked records and integrity hashes and can be successfully verified.	Supports audits and provides evident compliance evidence	Export should preserve integrity information.
FR-006	Non - Functional	The system shall support complex filtering across 10 million historical audit records in under 1 second.	High	Benchmarking tests shows query response time less than 1 sec under the defined workload	Ensure fast access to historical audit information during compliance officer investigation.	Given requirements.
FR-007	Non - Functional	The system must protect the audit records and exported records from unauthorized access.	High	The unauthorised users cannot modify the audit records. Modification attempts are detected from integrity verification.	Audit records contain sensitive and legal information.	Access control and integrity verification must be enforced.

Actors

1. Application / Enterprise System

Centralized Audit Trail Compliance Engine

NAME: SAGAR K **SRN**: PES2UG24CS424

2. Compliance Officer
3. Security Auditor

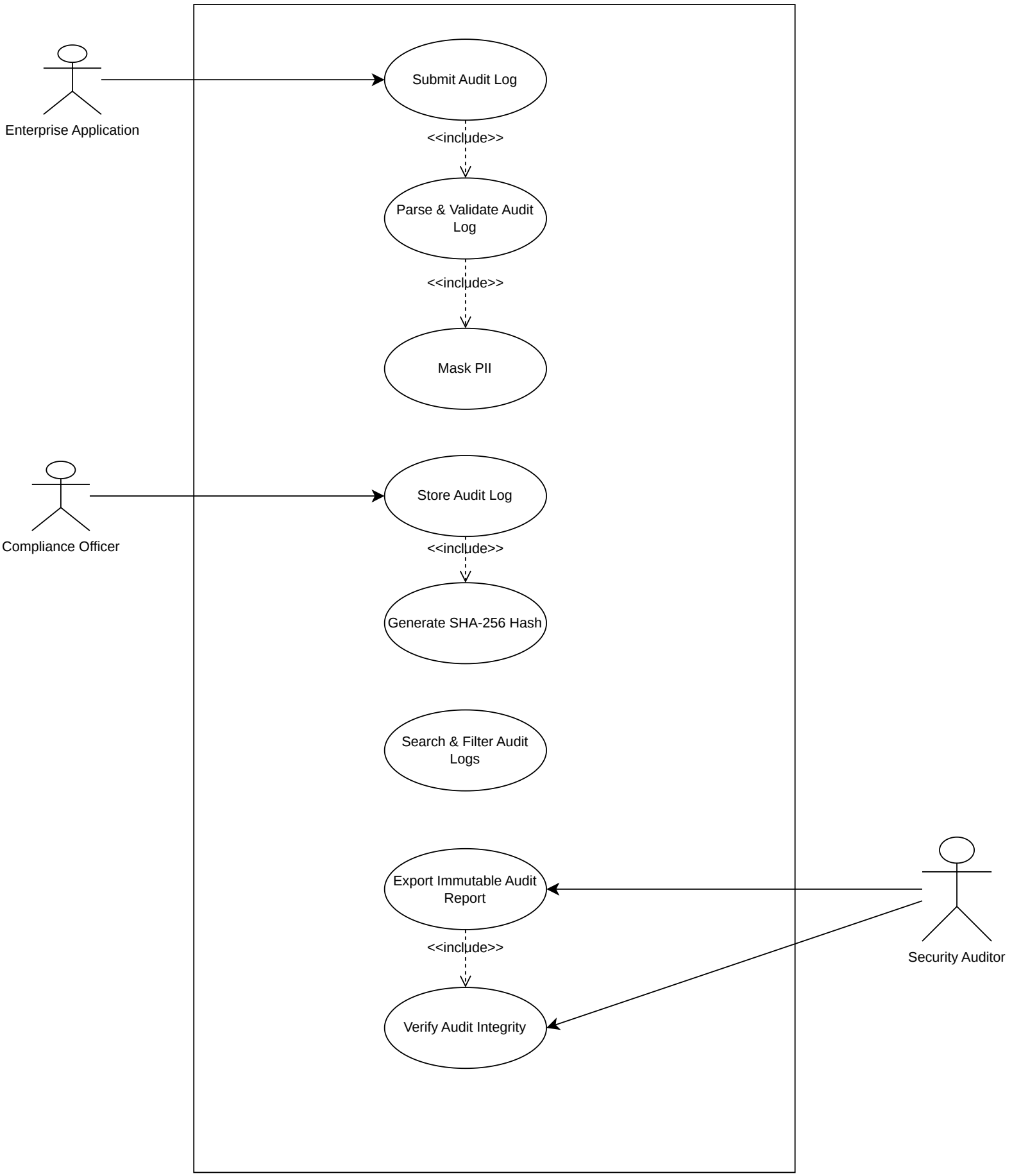
Main Use Cases

1. Submit Audit Log
2. Parse and Validate Audit Log
3. Mask PII
4. Store Audit Log
5. Generate Integrity Hash
6. Search and Filter Audit Logs
7. Export Immutable Audit Report
8. Verify Audit Integrity

Important UML Relationships

- Submit Audit Log «include» Parse and Validate Audit Log
- Parse and Validate Audit Log «include» Mask PII
- Store Audit Log «include» Generate Integrity Hash
- Export Immutable Audit Report «include» Verify Audit Integrity.

Centralized Audit Trail Compliance Engine



CENTRALISED TRAIL AUDIT COMPLIANCE ENGINE

NAME: SAGAR K **SRN: PES2UG24CS424**

Application Logs → Parse → Detect PII → Mask PII → Store Securely → Generate Hash → Auditor checks/export

Example:

Original:

Before Masking:

Credit Card = 4532-1234-5678-9012

After masking:

Credit Card = ****-****-****-9012

What these System actually do ?

- Generally companies / Apps will generate audit logs .
- This system collects audit logs from companies / User Application.
- Parse the audit logs
- It detects the personal information PII(like security number , Credit card number)
- It masks the personal information.
- It stores PII securely .
- Each time it generates hash SHA-256 for every audit logs.
- Compliance officer will check and filter the audit logs .
- It provides security , protection and reliability

Use-Case Flow: Submit and Process Audit Log

Use Case ID

UC-01

Use Case Name

Submit and Process Audit Log

Primary Actor

Application / Enterprise System

Supporting System

Centralized Audit Trail Compliance Engine

Goal

To receive an enterprise audit log, validate it, mask sensitive PII, generate an integrity hash, and securely store the audit record.

Preconditions

1. The audit logging system is available.
2. The enterprise application is authorized to submit audit logs.
3. The audit log is submitted in JSON format.

Main Success Scenario

1. The Enterprise Application submits a structured JSON audit log to the compliance engine.
2. The system receives the audit log.
3. The system validates the JSON structure and required fields.
4. The system identifies sensitive PII fields such as credit card numbers and Social Security Numbers.
5. The system masks all detected PII fields.
6. The system generates a SHA-256 integrity hash for the masked audit record.
7. The system stores the masked audit record and its integrity hash in secure storage.
8. The system returns a successful processing response to the Enterprise Application.
9. The use case ends successfully.

Alternate Flow

A1. Invalid Audit Log

1. At Step 3, the system detects that the submitted JSON is malformed or missing required fields.
2. The system rejects the audit log.
3. The system records the validation failure.
4. The system returns an error message to the Enterprise Application.
5. The invalid record is not stored as a valid audit record.
6. The use case ends.

A2. PII Detection Failure

1. At Step 4, the system cannot reliably process the submitted PII fields.

2. The system rejects the audit log instead of storing potentially sensitive cleartext data.
3. The system records the processing failure.
4. The system notifies the submitting application.
5. The use case ends unsuccessfully.

Postconditions

Success

The audit record is stored with PII masked and a valid SHA-256 integrity hash.

Failure

The invalid or unsafe audit log is not stored as a valid compliance record.